TASK REPORT 03 · QUANTITATIVE STRATEGY & ALPHA DISCOVERY

Quantitative Research Framework & Hypothesis Fleet

Complete Technical Record of the Macroeconomic Research Pivot, Point-In-Time Data Collectors, Qualification Standard v2, and Weak-Voters Committee (October 18 – 20, 2026)

RESEARCH SCHEMA	CARDS REGISTERED	BENCHMARK VERDICT	QUALIFIED SLEEVES
Postgres ext + research	23 Cards (11 Executed)	Legacy Indicators KILLED	3 Promoted to SHADOW

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1. The Strategic Research Pivot

On 2026-10-18, the quantitative engineering team completed a comprehensive audit of past trading performance and reached an incontrovertible conclusion: **single-pair technical indicator trading at retail spreads is mathematically exhausted and net negative-sum.**

The Benchmark Control Kill Test (2026-10-19)

To eliminate ambiguity before building new models, three classic indicator benchmarks were backtested over 16 years (2007–2022) with OANDA bid/ask replay and realistic friction (2.4 pip round trip + 0.6 pip slippage):

Strategy Key	Trades Evaluated	Net Expectancy	Total Losses	t-statistic	Verdict
`classic_ema_adx_trend` (H-01)	27,857 trades	-0.152 R / trade	-4,243.2 R	-20.60 (0/6 pos)	KILLED
`classic_donchian_breakout` (H-02)	55,652 trades	-0.137 R / trade	-7,606.4 R	-23.53 (0/6 pos)	KILLED
`classic_bollinger_reversion` (H-03)	50,761 trades	-0.126 R / trade	-6,407.6 R	-29.57 (0/6 pos)	KILLED

> **Conclusion:** Across over 134,000 backtested trades, classic technical indicators lost over 18,000 R. Optimizing moving average periods or RSI levels simply overfits random noise. Genuine edge must come from slow macroeconomic premia, institutional flows, or ensemble voting.

2. Point-In-Time (PIT) Data Collection Engine

To support fundamental and macroeconomic research under the strict **Free-Data-Only Rule**, automated collectors were built in `src/research/collectors`:

- **Look-Ahead Prevention:** Every data point is tagged with an `available_at` timestamp. Strategies access data strictly via `PitView(as_of)`, enforcing `available_at ≤ as_of`. Look-ahead bias is structurally impossible.
- **Macro Vintages:** Economic indicators subject to revision (CPI via ALFRED) store append-only revision vintages.
- **Active Collectors:**
  - `fred.py`: 10Y/2Y yields, Fed Funds, 5Y TIPS real yields, breakeven inflation, crude oil, S&P 500, VIX.
  - `yields.py`: Official Svensson spot curves from ECB, Bank of Canada, and Japan MoF.
  - `cftc.py`: Weekly CFTC Traders in Financial Futures (TFF) asset manager and leveraged money positions.
  - `policy_rates.py`: BIS daily policy rates covering all 8 major currency pairs from 2004+.
  - `fomc.py`: Federal Reserve FOMC announcement timestamps and statement text.
  - `financing.py`: Daily forward archive of real OANDA swap financing rates per pair.

3. Qualification Standard v2 & The PROMISING Tier

To eliminate data snooping, research is governed by an 8-step qualification framework and an append-only trial ledger in PostgreSQL (`research.trial_ledger`):

1. **Step 0: Pre-Registration.** Hypothesis card written, ≤ 2 free parameters, sha256 frozen in git.
2. **Step 1: In-Sample Backtest (2007–2022).** 6 temporal blocks with 5-day embargoes and conservative bid/ask replay.
3. **Step 2: Qualification Standard v2.**
   - `PROMISING TIER` : Net Mean $R > 0$, positive in ≥ 3 of 6 blocks, and $P(ext{edge} > 0) \ge 75\%$. Fast-tracks strategy to holdout testing.
   - `KILLED TIER` : Net Mean $R \le 0$ or negative in majority of blocks. Terminated permanently.
4. **Step 3: One-Shot Holdout (2023–2026).** Run exactly once. Failure kills immediately; passing earns promotion to DEV shadow forward paper testing.
5. **Deflated Sharpe Adjustment:** Backtest Sharpe ratios are discounted against effective trial counts ($K_{ ext{eff}}$) to mathematically prevent p-hacking.

4. The 3 Qualified Shadow Strategies

Out of 11 officially executed trials, 3 strategies cleared both in-sample vetting and the 2023–2026 one-shot holdout, earning promotion to **DEV SHADOW** forward testing:

Strategy Key / Card	Family / Style	Mechanism & Performance	Current Stage
`post_fomc_drift` **(H-20261018-06)**	Macro Event Swing (H1 Bars)	Trades USD pairs in direction of first-hour post-FOMC move, holding for 48 hours. • In-sample: +11.7 R **• 2023+ Holdout: +25.8 R (Sharpe 0.44, PF 1.38, Return/DD 2.14)**	DEV SHADOW
`vol_managed_carry` **(H-20261018-02)**	Macro Risk Premium (Daily Basket)	Cross-sectional carry basket across 7 pairs, de-risking when realized FX vol spikes. • In-sample: +7.3 R (4/6 blocks positive) **• 2023+ Holdout: +7.4 R (Annualized Sharpe +0.40 net of 1.1% fee drag)**	DEV SHADOW
`weak_voters_committee` **(H-20261019-04)**	Ensemble Condorcet (H4 Bars)	Combines 15 diverse weak directional voters (macro, flow, fractals, trend). Trades when consensus $\|A\| \ge 7$. • In-sample: +14.7 R ($P(ext{edge}) = 79.7\%$, 4/6 blocks pos) **• 2023+ Holdout: +1.6 R (4/6 blocks positive, t-stat 0.20)**	DEV SHADOW

> **Next Quantitative Milestone:** Implementing the A1 (`exit_by_utc`) and A2 (sleeve routing) contracts in System 2/3 so the Weak-Voters Committee and Post-FOMC Drift can execute paper trades directly through the live execution engine.